

Alpilla 600 MW Combined Cycle Power Plant

HSE Design Criteria

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A	IFR	2026-12-14	First issue for Owner approval (Basic Design Package)	EPCE	EPC HSE lead	EPC engineering manager

1. Purpose and scope

1.1 This document defines the HSE design criteria of the Alpilla 600 MW combined cycle plant: environmental limits, noise, occupational exposure, safety distances, hazardous area classification basis, escape routes and access, ergonomics, fire protection basis, the process safety study programme and the major accident basis. It is the basis for the HSE studies (HAZID, HAZOP, SIL, QRA, fire risk assessment, explosion protection document) and for the design of all disciplines.

1.2 The Contractor shall implement an ISO 45001-certified HSE management system and shall pursue a lost-time-injury target of zero (ER-14.05); the environmental management shall follow ISO 14001 as required by the Owner.

2. Governing requirements

ER-14.01: The Works shall comply with the conditions of the positive EIA decision and the Environmental Law (No. 2872) and its regulations.

ER-14.02: Natural gas: NO_x ≤ 30 mg/Nm³, CO ≤ 30 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. LDO: NO_x ≤ 50 mg/Nm³, CO ≤ 50 mg/Nm³, SO₂ ≤ 10 mg/Nm³, dust ≤ 5 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. All dry, 15 % O₂, 50 % to 100 % GT load (ALP-OWN-ENV-001 section 1).

ER-14.03: Maximum 85 dB(A) at 1 m from any equipment in normal operation.

ER-14.04: Maximum 45 dB(A) LAeq night at the nearest sensitive receptor.

ER-14.05: Construction and design shall comply with the Occupational Health and Safety Law (No. 6331); the Contractor shall implement an ISO 45001-certified HSE management system and a lost-time-injury target of zero.

ER-14.06: Hazardous area classification per IEC 60079-10-1; equipment certified accordingly.

ER-14.07: HAZID, HAZOP, SIL determination, fire and explosion risk assessment and a Major Accident Prevention assessment where applicable.

3. Codes and standards

3.1 The HSE design shall comply with the Turkish legislation on the environment (Law 2872, EIA Regulation, Environmental Permit Regulation, industrial air pollution and large combustion plant regulations, Water Pollution Control Regulation, Noise Regulation, Waste Regulation, Soil Regulation), on occupational safety (Law 6331, Construction OHS Regulation, Work Equipment Regulation, Explosive Atmospheres Regulation), on the prevention of major accidents (Major Accident Regulation), and on fire (Regulation on Fire Protection of Buildings), and with the international codes NFPA 850, NFPA 30, NFPA 11, NFPA 12, NFPA 20, NFPA 72, IEC 60079-10-1, IEC 61511 and EN 14181.

3.2 Codes named in this document but not in the register (EN 50272-2, NFPA 15, NFPA 24, NFPA 13, NFPA 101, EN 13463, IEC 60079-14, IEC 60079-17, ISO 9612, ISO 11690, the Turkish regulation on noise exposure at work, the Turkish regulation on chemical agents at work and the Turkish regulation on emergency response) shall be registered before they are cited in a study.

4. Environmental limits

Parameter	Value	Condition	Status	Basis
CO	30 mg/Nm3	dry, 15 % O ₂ , 50-100 % GT load	confirmed	ER-14.02
CO on LDO	50 mg/Nm3	dry, 15 % O ₂ , 70-100 % GT load	confirmed	SRC-OWNR-0007, ER-14.02
Dust on LDO	5 mg/Nm3	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
NH ₃ slip	5 mg/Nm3	dry, 15 % O ₂	confirmed	ER-14.02
NO _x (as NO ₂)	30 mg/Nm3	dry, 15 % O ₂ , 50-100 % GT load	confirmed	ER-14.02
NO _x on LDO	50 mg/Nm3	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
SO ₂ on LDO	10 mg/Nm3	dry, 15 % O ₂	confirmed	SRC-OWNR-0007, ER-14.02
Stack height	65 m	EIA decision (approved dispersion study)	confirmed	ER-06.06, SRC-OWNR-0007

4.1 Air emissions shall meet ER-14.02 and the EIA decision at the stack (65 m, EIA decision per the approved dispersion study); the values above apply to natural gas at the stated load range and to LDO firing for not more than 500 h/year; emissions shall be measured continuously by CEMS per EN 14181 (QAL1 certified per EN 15267-3, sampling location per EN 15259) with on-line reporting; the SCR ammonia slip shall be controlled at the stack to the limit above.

4.2 Water discharge: the temperature rise across the condenser shall not exceed 7 K in steady-state operation in all ambient conditions (TQ-041), the temperature excess at 100 m from the diffuser 3 K and the residual oxidant at the end of the outfall pipe 0.1 mg/l, sampled at the outfall seal pit without credit for mixing (DP-CASE-0015 to DP-CASE-0017, TQ-041); the electro-chlorination dosing shall follow DEC-EPCE-0015; process and sanitary effluents shall be treated to the sea-discharge limits of the Water Pollution Control Regulation before discharge to the CW outfall; the SWRO brine is returned only while a CW pump runs (DEC-EPCE-0015), so that the salinity excess at 50 m from the diffuser stays within DP-EPCE-0012.

4.3 Waste, soil and groundwater: all liquid storage areas (LDO tanks, transformer oil, chemicals, ammonia) shall have impervious bunds sized for the largest tank volume plus firewater and rainfall; wastes shall be segregated and managed per the Waste Management Plan (HS-PLN-0004); the soil and groundwater baseline (HS-RPT-0003) shall be established before construction so that any contamination can be attributed.

4.4 Marine and ecology conditions of the EIA: no marine works outside the window of DP-EPCE-0014; a fish return system at the screens with the intake velocity of DP-EPCE-0011; archaeological chance-find procedure for all excavations; construction noisy works only on weekdays 07:00 to 19:00 and Saturdays 08:00 to 17:00 with 7 days notice for steam blowing and silencers; traffic restrictions of the village road; each condition shall be entered in the EIA compliance matrix HS-RPT-0001 with the measure and the evidence document.

5. Noise

Parameter	Value	Condition	Status	Basis
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03
Noise at nearest receptor (day)	55 dB(A)	LAeq day	confirmed	SRC-OWNR-0007, ER-14.04
Noise at nearest receptor (night)	45 dB(A)	LAeq night	confirmed	ER-14.04

5.1 Equipment noise shall not exceed 85 dB(A) at 1 m from any equipment in normal operation (ER-14.03) and the plant noise shall not exceed 45 dB(A) LAeq at night (and the day limit DP-EPCE-0010 of the EIA) at the nearest sensitive receptor (ER-14.04); the requirement shall be verified by the noise study (HS-STU-0001), which shall use the sound power levels guaranteed by the suppliers, and by the noise survey after commissioning (ER-16.05).

5.2 Noise sources with a sound pressure level above 85 dB(A) at 1 m by the vendor data shall be fitted with acoustic insulation, silencers or enclosures by the vendor as part of the supply; noise levels in the requisitions shall be stated as sound power level and sound pressure level at 1 m; the noise limit shall be a guarantee with a test per the vendor.

5.3 Occupational noise: the upper exposure action value is 85 dB(A) (LEX,8h) and the peak limit 137 dB(C) (Turkish regulation on noise at work); areas with more than 85 dB(A) shall be marked as hearing protection zones, permanent workplaces (control room, offices, workshops) shall be designed for not more than 65 dB(A) where practicable (control room 40 dB(A), HVAC document); steam blowing, safety valve discharge and other intermittent noises shall use silencers.

6. Occupational exposure

6.1 Workers exposure to chemicals shall be kept below the occupational exposure limits of the Turkish regulation on chemical agents (which follow the EU indicative limits): ammonia 14 mg/m³ (20 ppm) as 8 hour average and 36 mg/m³ (50 engine technical_documents v1.2.0 | data df1f3745fae956df | generated 2026-09-29 - generated from the project database; do not edit

ppm) as 15 minute value; chlorine 1.5 mg/m³ (0.5 ppm) and 2.9 mg/m³ (1 ppm) as 15 minute value; nitrogen dioxide 0.96 mg/m³ (0.5 ppm) and 1.91 mg/m³ (1 ppm); hydrogen sulphide 5 ppm and 10 ppm; the design shall minimise release and shall provide the fixed gas detection of section 10.

6.2 Chemicals shall be stored and handled in closed systems where possible; the design shall provide safety showers and eye wash stations within 10 seconds (about 15 m) of every point where hazardous chemicals (ammonia, sodium hypochlorite, acids, caustic, hydrazine substitutes, LDO sampling) are handled, closed sample points, vent lines routed to safe locations, and no manual transfer of hazardous chemicals from drums without a closed transfer system.

6.3 Other physical agents: thermal insulation or personnel protection shall be provided for surfaces above 60 degC that can be touched from platforms and walkways (protection to a height of 2.1 m and 0.8 m from the edge); confined spaces (tanks, pits, the condenser water boxes, the CW tunnel) shall have permanent gas monitoring connections, two access points where practical and rescue provisions; lightning protection per IEC 62305 for all buildings and structures with the ground flash density of the site (DP-OWNR-0009).

7. Safety distances

7.1 Safety distances between hazardous installations and between hazardous installations and buildings, boundaries and ignition sources shall be established by the layout with the following minimum values until the QRA (HS-HAZ-0010) and the hazardous area classification (HS-HAZ-0008) determine other values: the minimum distances of NFPA 30 for the LDO tanks (2 x 3,800 m³, API 650); NFPA 850 for the oil-filled transformers (15 m to buildings and to other transformers or a fire barrier of 2 hour rating).

Item	Value	Unit	Status	Basis / remarks
Fuel gas receiving station and gas preheaters to occupied buildings and ignition sources	not less than 15 m; not less than 25 m to the plant boundary and public roads		assumption	ER-14.06, REF-CONS-0083; Confirm by QRA
LDO storage tanks to buildings, boundary and ignition sources	per NFPA 30, and not less than 15 m to buildings; dike volume per the drainage criteria		assumption	REF-CONS-0067; Tank farm bund and drainage
Aqueous ammonia storage (2 x 30 m ³) to air intakes, occupied buildings and boundary	not less than 20 m to occupied buildings and air intakes; not less than 30 m to the plant boundary		assumption	ER-14.07, ER-14.06; Inventory of about 54 t (TQ-034); confirm by dispersion in the QRA
Hydrogen sources (electrolysis hydrogen of the electro-chlorination, H ₂ / CO ₂ bottle racks, battery rooms) to buildings and ignition sources	not less than 8 m for the bottle racks (600 Nm ³), open-air location with a light roof; electro-chlorination rooms ventilated per the process design		assumption	REF-CONS-0083
Oil-filled transformers to buildings and between units	15 m or a firewall rated 2 hours		confirmed	REF-CONS-0069
Sodium hypochlorite and acid / caustic storage to ammonia storage	separate bunds; not less than 10 m or a wall to avoid mixing of the chemicals		assumption	ER-14.05
Emergency assembly points	two points upwind of the gas station and ammonia store outside the hazard ranges, located by the QRA		assumption	ER-14.07

8. Hazardous area classification

8.1 Hazardous areas shall be classified per IEC 60079-10-1 (ER-14.06) with the Explosive Atmospheres Regulation of Turkey and equipment selection per ATEX 2014/34/EU; the classification report and drawings (ALP-EPC-00000-HS-HAZ-0008) shall list every release source, its grade (continuous, primary, secondary), the ventilation degree and availability, the zone and the extent, and shall feed the hazardous area equipment register (EL-LST-0004) and the explosion protection document (HS-RPT-0005).

8.2 Classification basis: natural gas (methane 85 to 98.5 mol %, gas group IIA, temperature class T1) at the fuel gas station, GT gas skid interface and gas piping; hydrogen (gas group IIC, T1) at the hydrogen sources and the battery rooms (and up to 5 vol % hydrogen admixture in the fuel gas capability of DP-CASE-0021, which shall be assessed as a release of a group IIC constituent for the equipment selection at the gas station); LDO handled below its flash point (DP-EPCE-0013) is not classified except at vents of the tanks (zone 2 in the vent zone); aqueous ammonia 24.5 % (IF-25) is not an explosive atmosphere at ambient conditions unless heated.

8.3 Equipment in zones shall be certified (Ex d, Ex e, Ex i, Ex p, Ex n as appropriate, with at least IIB + H₂ (or IIC in hydrogen zones) and T3 minimum unless a lower temperature class is required by the substances); the minimum category is 2G for zone 1 and 3G for zone 2; outdoor equipment shall be IP66 in stainless steel or with a marine coating (C5); no non-Ex electrical equipment shall be installed in a classified zone; the electrical installation shall be per IEC 60079-14 and inspected per IEC 60079-17.

8.4 Non-hazardous rooms adjacent to classified areas (control building, electrical rooms, analyser shelters) shall be outside the zones with pressurisation or an air lock where required and shall be protected against gas ingress by the HVAC design (HV-DBR-0001, gas detection at intakes).

9. Escape routes, access and ergonomics

9.1 Every building, platform and process area shall have at least two independent escape routes leading in opposite directions where the occupancy allows, with a maximum travel distance to a place of safety of 45 m (30 m within the turbine halls, gas station and other classified areas) and a maximum dead end of 15 m; the escape routes shall have a minimum clear width of 0.9 m (1.2 m for main escape routes and stairs from the control and administration buildings) and a minimum clear height of 2.1 m and shall be kept free of equipment and laydown.

9.2 Platforms, stairs and ladders: stairs at not more than 45 degrees with a rise not more than 200 mm and a tread not less than 250 mm, handrails at 1.1 m with a mid rail and 100 mm toe plate, ladders with cages (or fall arrest rails) above 3 m and landings every 6 m; access to all valves, instruments, sample points and equipment requiring operation or maintenance more than once a year by permanent platforms or stairs, not by ladders; slip-resistant grating and no trip hazards.

9.3 Ergonomics: manually operated valves and controls with a handwheel centre between 0.6 m and 1.8 m above the working level (preferably 0.9 m to 1.5 m); local indicators readable between 1.0 m and 1.8 m; access clearance at least 0.8 m in front of operated equipment; the manual handling of loads above 25 kg shall be avoided by hoists, monorails or lifting lugs; equipment shall be laid out for maintenance with lay-down and lift paths agreed in the constructability and maintainability review (HS-RPT-0002).

9.4 Emergency lighting shall give at least 1 lux (at floor level) on escape routes and 5 lux at fire-fighting and first-aid equipment for 90 minutes (EL-DBR-0001 section 14) from the UPS / battery; escape and exit signs shall be photoluminescent or illuminated; the emergency response plan (HS-PLN-0005) shall define at least two assembly points, wind socks visible from all process areas, safety showers, first-aid stations and the fire station response.

9.5 Vehicle access: the fire brigade shall have access to all buildings and process areas from two directions at least, with the fire road width of 6 m and turning radius of 12.5 m, hydrants along the roads and clear access to the fire water tanks and pumps; the roads shall be designed per the civil design criteria (CV-DBR-0001).

10. Fire protection basis

10.1 Fire protection shall follow NFPA 850 for electric generating plants and the Fire Regulation, with fire protection features per NFPA 20 (pumps), NFPA 11 (foam), NFPA 12 (CO₂) and NFPA 72 (detection and alarm); the fire protection concept and the fire risk assessment (ALP-EPC-00000-HS-HAZ-0009) shall define the fire areas, the passive protection and the active systems and shall be approved before the issue of the fire water design.

10.2 Fire water shall be provided for the design flow and pressure of DP-EPCE-0006 (the LDO tank, being the largest single hazard, plus a hose stream allowance, at 110 m head) with the storage of DP-EPCE-0007 (2 x 100 % tanks, 2 hours); the fire water is fresh water filled from the SWRO permeate (DEC-EPCE-0018); the fire pumps are one electric and one diesel pump each of 100 % (NFPA 20, DEC-EPCE-0014), the diesel pump shall start automatically on the low pressure of the ring main and the electric pump shall be supplied from the essential section; the fire water main shall be a buried looped ring main sectionalised so that no single break cuts the supply to more than one fire area.

DEC-EPCE-0014 Preliminary equipment sizing basis and conventions: Equipment is sized from the heat and mass balances (all cases; the maximum case governs flows) with the margins of the process design criteria: pumps +10 % flow and +5-10 % head, motors to the next IEC standard size, motors of 200 kW and above on the 10.5 kV system (motor rated voltage 10 kV). The rated power of a package is the power of its own consumers that are not listed as separate tags (no double counting in the electrical load list). The CW system is designed for 38,000 m³/h (DEC-EPCE-0002; heat balance 36,050 m³/h). LDO storage 2 x 3,800 m³ nominal (3,500 m³ working each) gives 72.8 h at base load on LDO (80.7 t/h, 840 kg/m³). Fire protection per NFPA 850 / NFPA 20: fire water design flow 341 m³/h (1,500 gpm) at 110 m, electric and diesel fire pumps 1 x 100 % each, fire water tanks 2 x 100 % of 700 m³ (2 h). Natural gas preheating at the receiving station by 2 x 100 % indirect water-bath heaters (gas-fired, 2.2 MW thermal each) against Joule-Thomson cooling and to keep 28 K superheat at the GT skid; heater fuel is plant fuel and is included in the plant heat balance when the supply gas is below 20 degC.

10.3 Fire protection systems by hazard shall be: automatic foam (NFPA 11) and cooling water for the LDO tanks; deluge for oil-filled transformers (NFPA 15); water mist or CO₂ (NFPA 12) for the gas turbine enclosure and the lube oil areas; gaseous or water mist for the control room floor voids and the cable rooms; automatic sprinklers or hydrants and hose reels for the buildings as required by the Fire Regulation; portable extinguishers in all buildings; hydrants with a maximum spacing of 60 m.

10.4 Fire and gas detection: fire detection in all buildings and hazardous areas per NFPA 72 with addressable devices and a fire alarm panel in the control room and the fire station; combustible gas detectors (10 % LEL alarm, 20 % LEL high alarm, 40 % LEL trip of the gas supply where defined by the study) at the gas station, the GT enclosure inlets, the fuel gas skid, the hall ventilation intakes and the battery and hydrogen areas; ammonia detectors (25 ppm alarm, 50 ppm high) at the ammonia unloading and storage; the detector positions shall be based on the detector mapping study (HS-STU-0003).

10.5 Passive fire protection and compartmentation: the control and electrical building, the GIS building and the fire station shall be separate fire compartments with 2-hour walls; the cables shall be routed with fire barriers at compartment crossings; fire walls of transformers shall extend at least 0.6 m above and 0.6 m beyond the transformer and its conservator; structural fire resistance per the civil design criteria.

11. Process safety studies and major accident

11.1 The process safety studies shall be carried out by the Contractor as required by ER-14.07: the HAZID (HS-HAZ-0001) at the start of the basic design; the HAZOP of every system node (fuel gas and gas station HS-HAZ-0002, water-steam cycle HS-HAZ-0003, LDO and ammonia HS-HAZ-0004, water treatment and chemicals HS-HAZ-0005, BOP utilities HS-HAZ-0006) on the P&IDs at IFR status, with the close-out report (HS-RPT-0004) before the P&IDs are issued for construction; the SIL determination (HS-HAZ-0007) per IEC 61511; the fire and explosion risk assessment (HS-HAZ-0009) and the QRA (HS-HAZ-0010).

11.2 Risk acceptance criteria (project criteria, to be agreed with the Owner): individual risk of death for workers not more than 1×10^{-3} per year (intolerable above) and for members of the public not more than 1×10^{-4} per year, with risk reduced to as low as reasonably practicable and a broadly acceptable region below 1×10^{-6} per year; the societal risk shall be shown on the FN curve; a fire, explosion or toxic release shall not cause the escalation to a second major hazard within the first 30 minutes for the design events (LDO tank fire, gas release at the station, ammonia release).

11.3 Safety integrity: safety instrumented functions shall be classified by the SIL determination (LOPA or risk graph) with the target of the functions in the risk criteria; the SIL-rated functions (burner management, gas station shutdown, LDO trips, ammonia unloading interlock, fire and gas) shall be verified for the achieved PFDavg with the proof test intervals and shall not be less than SIL 1; the HAZOP recommendations and their close-out shall be tracked in the HSE-in-design register (HS-PLN-0001).

11.4 Major accident: the plant is a lower-tier establishment under the Major Accident Regulation because the LDO inventory (2 tanks per DEC-EPCE-0014) exceeds the lower-tier threshold of 2,500 t for petroleum products (notification PMT-015). The other dangerous substances (natural gas at the gas station, hydrogen, aqueous ammonia 24.5 % of about 54 t of solution per TQ-034, which is not 'ammonia, anhydrous' and counts under its hazard categories) are included in the inventory and the aggregation rule. The Major Accident Prevention Policy and the notification are prepared per HS-HAZ-0010 and updated when an inventory changes.

11.5 Explosion protection document: an explosion protection document per the Explosive Atmospheres Regulation shall be prepared for the plant (HS-RPT-0005), consolidating the hazardous area classification, the equipment and the organisational measures; the design of buildings with a gas explosion risk shall include the explosion venting or blast loads as determined by the study.

12. EIA compliance and HSE in design

12.1 Every condition of the positive EIA decision (SRC-OWNR-0007) shall be traced to the design measure and evidence in the EIA Conditions Compliance Matrix ALP-EPC-00000-HS-RPT-0001, which shall be updated at each design issue and shall be the record of compliance with ER-14.01; open conditions shall be raised as TQ or design actions.

12.2 HSE in design: safety-in-design reviews shall be held at the end of the basic design (constructability and maintainability, HS-RPT-0002) and at 60 % of the detailed design (3D model review) using the HSE-in-design register (HS-PLN-0001) with the actions closed before IFC; the occupational health and safety risk assessment (HS-RPT-0006) and the construction HSE plan (HS-PLN-0002) shall use the design hazards identified; the construction and environmental management plan shall follow HS-PLN-0003.

12.3 Future carbon capture: the space reservation and the tie-in points of ER-14.08 shall be included in the HAZID, the layout risk assessment and the safety distances, so that the future plant does not reduce the safety distances of section 7.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Law 2872	Environmental Law (Cevre Kanunu)
EIA Regulation	Environmental Impact Assessment Regulation (CED Yonetmeligi)
Env. Permit Regulation	Environmental Permit and Licence Regulation (Cevre Izin ve Lisans Yonetmeligi)
SKHKKY	Industrial Air Pollution Control Regulation (Sanayi Kaynakli Hava Kirliliginin Kontrolu Yonetmeligi)
LCP Regulation	Large Combustion Plants Regulation (Buyuk Yakma Tesisleri Yonetmeligi)
WPCR	Water Pollution Control Regulation (Su Kirliligi Kontrolu Yonetmeligi)
Noise Regulation	Environmental Noise Control Regulation (Cevresel Gurultunun Kontrolu Yonetmeligi)
Waste Regulation	Waste Management Regulation (Atik Yonetimi Yonetmeligi)
Soil Regulation	Soil Pollution Control and Point-Source Contaminated Sites Regulation
Major Accident Regulation	Regulation on Prevention and Mitigation of Major Industrial Accidents (Seveso III)
Law 6331	Occupational Health and Safety Law (Is Sagligi ve Guvenligi Kanunu)
Construction OHS Regulation	Regulation on OHS in Construction Works (Yapi Islerinde ISG Yonetmeligi)
Work Equipment Regulation	Regulation on Health and Safety in the Use of Work Equipment
Explosive Atmospheres Regulation	Regulation on Protection of Workers from Explosive Atmospheres
Fire Regulation	Regulation on Fire Protection of Buildings (Binalarin Yangindan Korunmasi Hakkinda Yonetmelik)
Machinery Directive 2006/42/EC	Machinery Directive, transposed as the Machinery Safety Regulation (Makine Emniyeti Yonetmeligi)
ATEX 2014/34/EU	ATEX Equipment Directive, transposed as the regulation on equipment for potentially explosive atmospheres
NFPA 12	Standard on Carbon Dioxide Extinguishing Systems
NFPA 85	Boiler and Combustion Systems Hazards Code
API 650	Welded Tanks for Oil Storage
NFPA 30	Flammable and Combustible Liquids Code
NFPA 11	Standard for Low-, Medium-, and High-Expansion Foam
NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and HVDC Converter Stations
NFPA 20	Standard for the Installation of Stationary Pumps for Fire Protection
NFPA 72	National Fire Alarm and Signaling Code
IEC 62305	Protection against lightning (all parts)
IEC 61511	Functional safety - Safety instrumented systems for the process industry sector
IEC 60079-10-1	Explosive atmospheres - Classification of areas - Explosive gas atmospheres
EN 14181	Stationary source emissions - Quality assurance of automated measuring systems
EN 15267-3	Air quality - Certification of automated measuring systems - Part 3
EN 15259	Air quality - Measurement of stationary source emissions - Measurement sections and sites
ISO 45001	Occupational health and safety management systems
ISO 14001	Environmental management systems - Requirements